Syed Hussam Abbas

Plant Molecular Biologist | RNA Surveillance (NMD), Heat Stress Recovery and Thermomemory

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RESEARCH INTERESTS

Post-transcriptional regulation of gene expression in plants, with a focus on nonsense-mediated mRNA decay (NMD), alternative splicing, heat shock factor (HSF) biology, and thermomemory in *Arabidopsis thaliana*; building on earlier work on transcription fidelity and alternative splicing control. My broader interest is in the productive transcriptome, the fraction of transcripts that escape surveillance and are actually translated, and in how post-transcriptional and translational quality control jointly set it during stress exposure and recovery. A current direction is to treat RNA surveillance capacity as a quantitative trait that may vary between Arabidopsis accessions and shape thermal acclimation. I am also interested in how plants sense environmental signals through phytochromes, cryptochromes and hormone signaling, and in the proteomics of the protein complexes.

EDUCATION

PhD in Plant Sciences Oct 2022 – Mar 2027

Hungarian University of Agriculture and Life Sciences (MATE), Gödöllő, Hungary

Supervisor: Dr. Tibor Csorba, Plant Stress Biology Group

Thesis: "Post-transcriptional regulation of heat stress response and thermomemory by nonsense-mediated decay in *Arabidopsis thaliana*"

- Showed that NMD sets the balance between HSFA2 splice isoforms during heat stress acclimation and acts as a pro-recovery factor after heat stress, independently of HsfA2 and HsfA3
- Contributed to work on transcription fidelity and co-transcriptional splicing control under heat stress, defining a cooperative TFIIS/NMD pathway for transcriptome quality control
- Established heat stress phenotyping regimes for acclimation, recovery and memory scoring, now used as standard protocols in the group

MSc in Plant Breeding and Genetics Sep 2019 – Jul 2022

University of Agriculture Faisalabad, Pakistan

Thesis: "Assessment of association among different yield contributing factors in maize (*Zea mays* L.) germplasm"

- Quantitative genetic analysis of trait associations across a maize germplasm panel, including correlation and path coefficient analysis of yield components

RESEARCH EXPERIENCE

Doctoral Researcher Oct 2022 – Present

Plant Stress Biology Group, MATE University, Gödöllő, Hungary

- Investigated NMD pathway regulation of HSFA2 alternative splicing isoforms (HSFA2-I and HSFA2-II) during heat stress acclimation and thermomemory using *upf* mutant lines
- Designed, executed and optimized controlled heat stress, acclimation and thermomemory phenotyping assays, establishing standardized acclimation, recovery and memory-scoring protocols across multiple *Arabidopsis* genotypes
- Performed isoform-specific RT-qPCR to dissect temperature-dependent splicing outcomes and NMD-sensitive transcript dynamics
- Conducted western blotting and SDS-PAGE for protein-level validation of stress response phenotypes

- Applied FPLC-based protein purification for biochemical characterization of stress-related protein complexes
- Contributed RT-qPCR, genotyping, and western blotting to a co-authored study on TFIIS/NMD cooperative transcriptome quality control under heat stress (Szaker et al., *Plant Cell*, 2025)
- Performed MPA treatment experiments and confocal/fluorescence microscopy for cellular phenotyping
- Conducted bioinformatics analyses including RNA-seq data interpretation and multi-omics dataset integration

Research Assistant Sep 2020 – Apr 2022

University of Agriculture Faisalabad, Pakistan

- Supported molecular biology experiments including nucleic acid extraction, PCR-based genotyping, and sample preparation
- Contributed to data collection, experimental execution, and results interpretation across active research projects

Research Intern Feb 2019 – May 2019

Ayub Agricultural Research Institute, Faisalabad, Pakistan

- Assisted in laboratory and field-based experimental work; supported data recording and research activities

PUBLICATIONS

1. **Abbas S. H.**, Szaker H. M., Darkó É., Erős T., Silhavy D. and Csorba T. (2026) Nonsense-Mediated mRNA Decay ambivalently regulates heat stress memory. *Manuscript submitted.*

2. Szaker H. M.*, Verma R.*, Szádeczky-Kardoss I., Gál N., **Abbas S. H.**, Darkó É., Pettkó-Szandtner A., Silhavy D. and Csorba T. (2025) Transcription fidelity and control of alternative splicing contribute to heat stress survival in Arabidopsis. ***The Plant Cell**, 37(11): koaf256.* https://doi.org/10.1093/plcell/koaf256

3. Nyikó T., Gyula P., Ráth S., Sós-Hegedűs A., Csorba T., **Abbas S. H.**, Bóka K., Pettkó-Szandtner A., Móricz Á. M., Molnár B. P., Erdei A. L. and Szittya G. (2025) INCREASED DNA METHYLATION 3 forms a potential chromatin remodelling complex with HAIRPLUS to regulate DNA methylation and trichome development in tomato. ***The Plant Journal**, 121: e70085*. https://doi.org/10.1111/tpj.70085

4. Singh K., **Abbas S. H.**, Singh S., Iqbal N., Kocsy G., Pérez-Pérez J. M., Riyazuddin R. (2025) Deciphering the Plants' Response to Flood-Induced Hypoxia and Anoxia at Molecular Level. In: Riyazuddin R., Gupta R., Ramteke P. W., Sayyed R. (eds) Plant Flooding. Plant in Challenging Environments. Springer, Cham, 6:93-112. https://doi.org/10.1007/978-3-031-83068-6_5

Full list of publications can be found here: https://syedhussamabbas.github.io/#publications

TECHNICAL SKILLS

RNA & Molecular Biology

- RT-qPCR / isoform-specific qPCR, PCR, molecular cloning, DNA/RNA extraction, plasmid preparation
- Alternative splicing analysis and quantification of NMD-sensitive transcript isoforms
- MPA treatment
- Design of heat stress assays for short- and long-acquired thermotolerance (SAT, LAT, TMHT), heat stress acclimation, recovery and thermomemory

Nuclear Biology & Chromatin

- Nuclei isolation from sample tissues
- Nuclear protein extraction and sample preparation for proteomics
- Histone extraction for post-translational modification analysis

Protein Biochemistry

- Western blotting, SDS-PAGE, co-immunoprecipitation (Co-IP)
- Recombinant protein purification by FPLC (affinity/size-exclusion)
- Biochemical characterization of protein complexes
- Subcellular fractionation (nuclear and cytoplasmic)

Microscopy & Imaging

- Fluorescence microscopy (Leica) and confocal laser scanning microscopy
- Subcellular localization of fluorescent protein fusions
- Optical (bright-field) microscopy

Bioinformatics & Data Analysis

- RNA-seq data interpretation, differential expression and splicing analysis
- R programming for statistical analysis and data visualization
- Familiarity with plant genome databases (TAIR, Ensembl Plants)

Plant Biology

- *Arabidopsis thaliana* cultivation, transformation, genotyping and phenotyping
- Controlled abiotic stress experiments (heat, hypoxia/anoxia) in growth chambers
- Genotyping of T-DNA insertion and EMS mutant lines

ADDITIONAL EXPERIENCE

Campus Mentor Jul 2024 – Jun 2026

Stipendium Hungaricum Mentor Network (HOOK), Hungary

- Mentored international PhD and MSc students on academic integration, administrative procedures, and well-being
- Coordinated student activities and prepared structured monthly progress reports

R&D Associate Jun 2022 – Sep 2022

BioTech Life Sciences Pvt Ltd, Multan, Pakistan

- Assisted in laboratory experiments and product-oriented R&D; supported data documentation and analysis

AWARDS & SCHOLARSHIPS

- Best Poster Award (de.NBI 2026: Bioinformatics for Proteomics, RUB Bochum, Germany)
- SH Dissertation Scholarship (Hungarian government scholarship for international doctoral researchers)
- Best Publication Award (GBI Day 2025, UniMATE, Hungary)
- Best Poster Award (E^3UDRES^2 Science Festival 2025, Latvia)
- Stipendium Hungaricum Scholarship (Hungarian government scholarship for intl doctoral researchers)
- Pannónia Mobility Scholarship (Hungarian mobility award for international research activities)
- Chief Minister's Merit Scholarship (competitive academic merit award, Pakistan)

REFERENCES

Available upon request.